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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Monke
Project / Solution Name	Buns at Work, Perishable Food Inventory Forecaster and AI Recommender System
Name (Leader)	Sean Abraham Summertan
Email & Phone (Leader)	s34n.4.50@gmail.com & +60 17-822-1132
University Affiliation(s)	Xiamen University of Malaysia
Live Prototype URL	https://sdc-hackathon--polychch3.replit.app https://sdc-dashboard.onrender.com/
Code Repository URL (public / access granted to judges)	https://github.com/chuch3/sdc-hackathon

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Buns at Work! connects end-to-end through a lightweight, cloud-hosted architecture. The user interface is a Streamlit dashboard where the bakery manager uploads messy sales CSVs, views the daily action list, and approves, edits, or rejects recommendations. This frontend communicates with a FastAPI backend that acts as the central router, enforcing the data-trust gate and contract-first logic. The backend reads and writes to a SQLite database via SQLAlchemy, which stores the First-Expired-First-Out (FEFO) inventory ledger, batch expiry dates, and the immutable audit log of every human approval. For AI inference, the backend first calls its local rule-based forecaster as the reliable default; if deeper demand modelling is needed, it routes the request to an optional PyTorch ML challenger hosted on a RunPod serverless GPU endpoint. If RunPod is unavailable or confidence is low, the backend automatically falls back to the baseline forecaster, ensuring no recommendation ever fails. The entire application is hosted and deployed on Replit, which runs both the Streamlit frontend and FastAPI backend in the cloud. A simple system block diagram is attached in the appendix showing the flow from UI → FastAPI → SQLite → RunPod, with the fallback loop clearly marked.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python	Primary programming language for the backend, forecasting engine, scripts, and dashboard.	PSF License (Python Software Foundation License)
FastAPI	REST API framework; serves HTTP/JSON endpoints between the Streamlit dashboard and the service layer. Enables backend and UI work in parallel. Provides the API tester at `/docs`.	MIT License

Name & Version	Purpose in Project	License
Streamlit	Dashboard framework; renders the UI (inventory view, expiry alerts, recommendations, bakery planner, KPI cards/charts, SDG impact dashboard, weekly quest board).	Apache License 2.0
SQLAlchemy	ORM mapping the data model (products, suppliers, batches, stock_movements, purchase_orders, etc.) to the database, allows SQLite for the demo with PostgreSQL as a later configuration change.	MIT License
SQLite	Database used for the demo/MVP (single location, synthetic seed data).	Public Domain
pytest	Test suite framework; unit tests, E2E tests, regression tests. A pytest suite is a stated deliverable.	MIT License
Ruff	Linter/formatter listed among the core stack for code quality.	Composite (MIT AND 0BSD AND Apache-2.0 AND others)
PyTorch	Used only for the optional ML challenger forecast (predicts daily q10/q50/q90 demand with one global model and product/category embeddings).	BSD-3-Clause (with additional attribution clauses)
MLflow	Experiment tracking and model management for the ML challenger, training, MLflow, and the model card live in `forecast/ml/` and `forecast/docs/`.	Apache License 2.0
Alembic	Database migration tool used in the run command `uv run python -m alembic upgrade head` before seeding.	MIT License

Name & Version	Purpose in Project	License
Docker	Containerizes and runs the whole stack (API + dashboard), `docker compose -f backend/compose.yaml up -d --build`.	Apache License 2.0
uv	Python package/project manager used in the run commands to execute Alembic, seeding, the API start script, and Streamlit.	Dual: Apache-2.0 OR MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
ASCII Magic	Decorative animated ASCII background	Free
Orbyte	Optional HR/leave portal link and navigation only with no outbound API call	Sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Render	Runs the core business logic for FEFO inventory, expiry alerts, demand forecasting, buy/bake recommendations, SDG metrics. Exposes REST endpoints consumed by the dashboard. For the dashboard, it deploys the UI inventory view, expiry alerts, recommendation cards, bakery bake plan, KPI charts, SDG impact dashboard. Calls the FreshStock API over HTTP.	Free / Singapore

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Quantile DNN	PyTorch	Self-trained	Apache License 2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo Dataset	docs/samples	Synthetic	Apache License 2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✓ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenCode Zen	AI model gateway providing access to free, OpenAI-compatible models for code generation and assistance during the build.	Free
OpenAI Codex	AI coding agent used to write, review, debug, and assist with software development tasks (e.g., generating code for the FastAPI backend or Streamlit dashboard).	Free

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-generated output is treated as a draft, not a final product. A qualified team member reviews it for accuracy, completeness, tone, and fitness for purpose before it goes anywhere. Reviews are documented so there's a clear record of who checked what and when. Outputs are validated against defined criteria then factual verification against trusted sources, spot-checks on calculations or code, and edge-case testing where applicable. For recurring use cases, we run sample batches and compare results against known-good baselines before scaling up.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✓ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core	Hosting the FastAPI + Streamlit app as a deployed web service, replacing local Docker.
Runpod	Compute	Hosting the FastAPI + Streamlit app as a deployed web service, replacing local Docker.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor resources were not essential to the core functionality of the prototype and could easily be replaced. The application is designed to run entirely on a local machine using Docker Compose or simple command-line scripts, with no dependency on Replit's hosting, database, or authentication services.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application itself does not collect, store, or send any user or personal data to third-party services. It runs locally, uses synthetic seed data, and explicitly excludes authentication, so there are no user accounts or personal information involved. No user data is collected or sent by the running prototype itself.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The main technical limitation of the AI component is forecast unreliability. The optional ML challenger could produce poor or unfounded demand predictions, so it is kept behind a rule-based baseline and must fall back to it whenever data is missing, stale, or low-coverage. It is only promoted if WAPE does not regress, bias stays within 10%, and interval coverage is near 80%. The system also shows forecast confidence and measures bias on a holdout period, uses fallback defaults and manual overrides for sparse or censored demand, and never lets ML output bypass human approval. SDG impact is labeled as estimates with assumptions shown. For the development tools, OpenCode Zen's free tier has rate limits and rotating models, and Codex depends on a single provider, but the repository does not document a formal mitigation for these. In practice the team would switch models or code manually.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24/10/2026